



IIE 8560

PROBABILISTIC MODELING AND BAYESIAN INFERENCE

Lecture 2: Bayesian Statistics

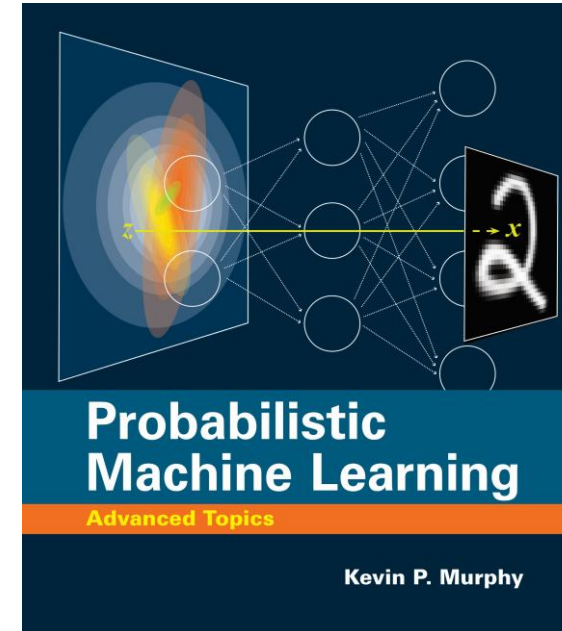
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Chapter 3

Bayesian Basics

Why Estimate Parameters?

- Probabilistic models depend on parameters θ which control the behavior of the distribution $p(x|\theta)$.
- Real-world applications require **learning θ from data** $\mathcal{D} = \{x_1, \dots, x_N\}$.
- Two paradigms for learning:
 - Frequentist: θ is fixed but unknown — learn via data only.
 - Bayesian: θ is a random variable with a prior — update beliefs with data.

"We usually assume the data are generated i.i.d. from $p(x|\theta)$, and we wish to learn θ ."

(Murphy, 3.2)

Bayesian Inference

- **Goal:** Infer unknown parameters θ using observed data $\mathcal{D}$.
- **Bayes' rule:**

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

- **Terminology:**
 - $p(\theta)$: Prior
 - $p(\mathcal{D} \mid \theta)$: Likelihood
 - $p(\theta \mid \mathcal{D})$: Posterior
 - $p(\mathcal{D})$: Evidence or marginal likelihood

Bayesian Estimation Framework

- Bayesian inference introduces a **prior** on θ :

$$p(\theta)$$

- Then updates this with data using Bayes' rule:

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

- Posterior represents updated beliefs.
- Predictive distribution incorporates this uncertainty:

$$p(x^*|\mathcal{D}) = \int p(x^*|\theta)p(\theta|\mathcal{D})d\theta \quad \rightarrow \text{Theta에 대한 marginalization}$$

- Bayesian estimation = uncertainty-aware learning.

Coin Toss Example (Beta-Bernoulli Model)

- Model:

$$x_n \sim \text{Bernoulli}(\theta)$$

$$\theta \sim \text{Beta}(a, b)$$

- Posterior:

$$\theta \mid \mathcal{D} \sim \text{Beta}(a + S, b + F)$$

$$\text{where, } S = \sum x_n, F = N - S$$

- Conjugacy makes posterior tractable.

Predictive Distribution (Posterior Predictive)

- Goal: Predict future data x_{N+1} .
- Integrate over posterior:

$$p(x_{N+1} = 1 \mid \mathcal{D}) = \mathbb{E}[\theta \mid \mathcal{D}] = \frac{a + S}{a + b + N}$$

- For $x_{N+1} \sim \text{Bernoulli}(\theta)$, predictive is a Beta-Binomial.

Posterior Predictive for Beta-Binomial

- Prior: $\theta \sim \text{Beta}(a, b)$
- Likelihood: $x_n \mid \theta \sim \text{Bernoulli}(\theta)$
- Posterior predictive:

$$p(x_{N+1} = 1 \mid \mathcal{D}) = \frac{a + S}{a + b + N}$$

$$p(x_{N+1} = 0 \mid \mathcal{D}) = \frac{b + F}{a + b + N}$$

- This forms the **Beta-Binomial** distribution.

Exchangeability Concept

- Informally: Order of observations does not matter.
- Formally:

$$p(x_1, x_2, \dots, x_N) = p(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}) \forall \text{ permutations } \pi$$

- Exchangeability $\Rightarrow$ existence of latent variable θ such that:

$$x_n \sim \text{iid given } \theta$$

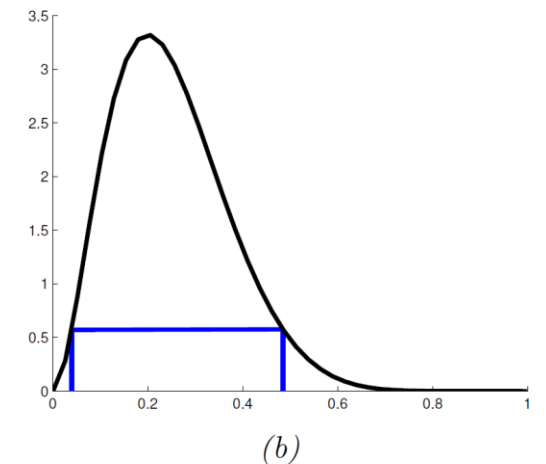
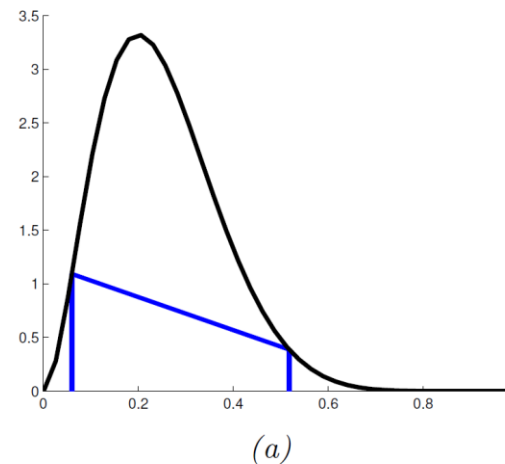
iid면 exchangeability (O)
exchangeability는 있어도 independence는 없을수도!
-> iid가 더 좁은 범위

Posterior Summary - Credible Intervals

- A **credible interval** is a Bayesian analogue to confidence intervals.
- Definition:

$$P(\theta \in [a, b] \mid \mathcal{D}) = 1 - \alpha$$

- Example: For $\theta \sim \text{Beta}(\alpha, \beta)$, 95% CI can be computed from quantiles.
- Types:
 - Equal-tailed interval
 - Highest posterior density (HPD) interval



Frequentist Statistics

What is Frequentist Statistics?

- Frequentist statistics interprets **probability** as the **long-run frequency** of events.
- Model:
 - Parameters θ are fixed but unknown.
 - Data x are random because they result from a stochastic process.
- Inference is made **only from the data**, without specifying a prior distribution over θ .
- Focus is on methods like:
 - Point estimation (e.g., MLE)
 - Interval estimation (e.g., confidence intervals)
 - Hypothesis testing
- **The parameter does not have a probability distribution under the frequentist view.**

Properties of MLE

여기서부터 안들음 ㅋㅋ

- **Consistency:**

- As $N \rightarrow \infty$, $\hat{\theta}_{\text{MLE}} \rightarrow \theta^*$ in probability.

- **Asymptotic Normality:**

- $\hat{\theta}_{\text{MLE}} \sim \mathcal{N}(\theta^*, I^{-1}(\theta^*))$ as $N \rightarrow \infty$ cf) $V(\theta) \geq \frac{1}{I(\theta)}$

- **Efficiency:**

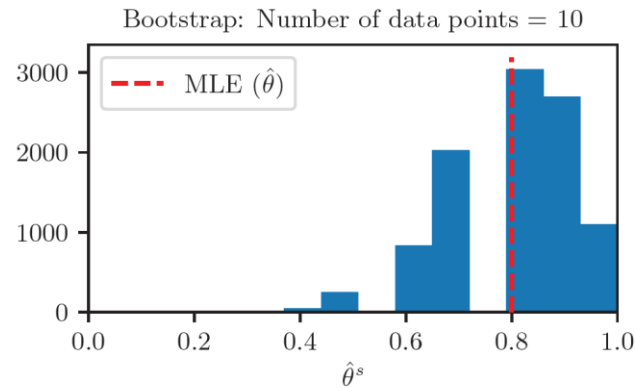
- Achieves the Cramér-Rao lower bound under regularity conditions.

- **Bias:**

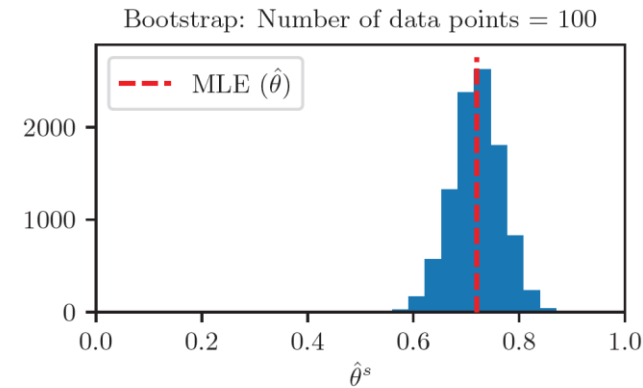
- MLE can be biased in finite samples.

Bootstrap approximation of the sampling distribution

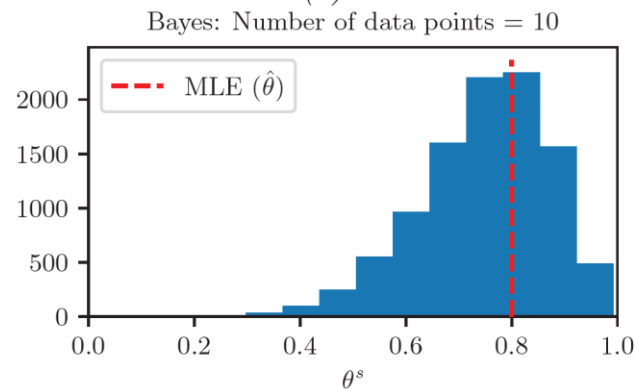
- Where the estimator is a complex function of the data, or when the sample size is small
 - approximate its sampling distribution using a Monte Carlo technique known as the bootstrap



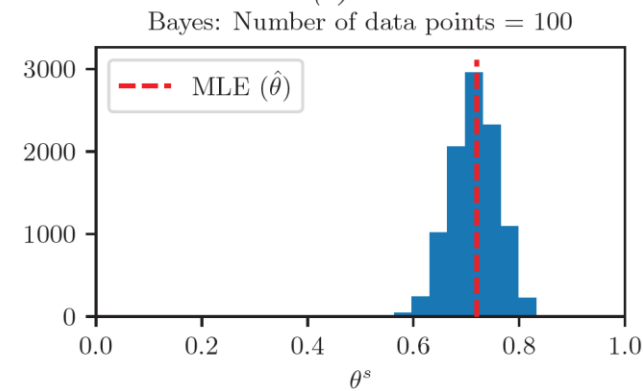
(a)



(b)



(c)



(d)

Likelihood and Log-Likelihood

- The likelihood function is:

Independent 가정 무조건 필요

$$\mathcal{L}(\theta) = p(\mathcal{D}|\theta) = \prod_{n=1}^N p(x_n|\theta)$$

- It's more convenient to work with the **log-likelihood**: 우리는 theta가 찾고 싶은것

$$\ell(\theta) = \log \mathcal{L}(\theta) = \sum_{n=1}^N \log p(x_n|\theta)$$

- Example: For Gaussian model with known variance:

$$\ell(\mu) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2$$

- **Goal of MLE:** Maximize this function to find the best θ .

Maximum Likelihood Estimation (MLE)

- MLE chooses parameters that maximize the likelihood:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D}|\theta)$$

- Example (again):

$$\hat{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{n=1}^N x_n$$

- Properties of MLE:
 - Consistent (as $N \rightarrow \infty$)
 - Asymptotically normal
 - Efficient (achieves Cramér-Rao lower bound)
- But: MLE can overfit when N is small or the model is too flexible.

Confidence Intervals

- A **confidence interval (CI)** provides a range that contains the true parameter value with a certain probability **over repeated sampling**.
- Example: 95% CI for θ means:
 - If we repeat the experiment many times, 95% of such intervals will contain θ .
- **Important:** This is not a Bayesian credible interval.
- For large N :
$$\theta \in [\hat{\theta} \pm 1.96 \cdot \text{SE}(\hat{\theta})]$$
- Where SE is the standard error.

Wald Confidence Intervals

- Use asymptotic normality:

$$\hat{\theta} \sim \mathcal{N}(\theta, \text{Var}(\hat{\theta}))$$

- Wald CI:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \sqrt{\widehat{\text{Var}}(\hat{\theta})}$$

- Valid if sample size is large and likelihood is approximately Gaussian near $\hat{\theta}$.
- Not reliable if model is mis-specified or sample size is small.

Fisher Information

- Measures the amount of information that the data carry about θ :

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log p(x | \theta) \right]$$

- Plays a role in:
 - Variance approximation for MLE
 - Hypothesis testing
 - Confidence interval construction

P-values in Frequentist Statistics

- A **p-value** is the probability of observing data **as extreme or more extreme** under the null hypothesis.

$$\text{p-value} = P(T(X) \geq T(x_{\text{obs}}) \mid H_0)$$

- Measures evidence **against** H_0 .
- Common threshold: 0.05
- A small p-value means **data is unlikely** under H_0 , not that H_0 is false.
- Bayesian methods do not rely on p-values; they use posterior distributions to assess evidence.

Conjugate Priors

What Are Conjugate Priors?

- A **conjugate prior** is one where the posterior is in the **same family** as the prior.
- Formally, if the likelihood is from an exponential family:

$$p(x|\theta) = h(x)\exp\{\eta(\theta)^\top T(x) - A(\theta)\}$$

- and the prior is:

$$p(\theta) \propto \exp\{\eta(\theta)^\top \tau - \alpha A(\theta)\},$$

- then the posterior becomes:

$$p(\theta|\mathcal{D}) \propto \exp\{\eta(\theta)^\top (\tau + \sum T(x_i)) - (\alpha + N)A(\theta)\}.$$

- Result: posterior is in the **same exponential family** as prior.

What Are Conjugate Priors?

- A prior distribution is **conjugate** to a likelihood function if the resulting posterior is in the same family as the prior.
- If $p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta) \cdot p(\theta)$ and $p(\theta|\mathcal{D}) \in \mathcal{F}$ whenever $p(\theta) \in \mathcal{F}$, then $\mathcal{F}$ is a **conjugate family**.
- Conjugate priors simplify the computation of posteriors:
 - Often yields closed-form solutions.
 - Avoids need for approximation methods like MCMC.
- The term "conjugate" was introduced by Raiffa and Schlaifer (1961).
- "A conjugate prior is one that, when combined with a given likelihood, yields a posterior of the same functional form as the prior."

Binomial-Beta Conjugacy

- **Likelihood:** $x \sim \text{Binomial}(N, \theta)$
- **Prior:** $\theta \sim \text{Beta}(a, b)$
- **Posterior:**
- $\theta|x \sim \text{Beta}(a + x, b + N - x)$
- **Intuitive interpretation:**
 - $a - 1$ and $b - 1$ represent prior pseudo-counts of success/failure.
 - Posterior adds observed data to prior pseudo-counts.
- **Example:** - $N = 10, x = 7$, prior = $\text{Beta}(1,1)$

→ posterior mean:

$$\mathbb{E}[\theta|x] = \frac{a + x}{a + b + N}$$

Multinomial-Dirichlet Conjugacy

- Likelihood:

$$\mathbf{x} \sim \text{Multinomial}(N, \boldsymbol{\theta})$$

- Prior:

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

- Posterior:

$$\boldsymbol{\theta}|\mathcal{D} \sim \text{Dirichlet}(\alpha_1 + n_1, \dots, \alpha_K + n_K)$$

- Posterior mean:

$$\mathbb{E}[\theta_k|\mathcal{D}] = \frac{\alpha_k + n_k}{\sum_i (\alpha_i + n_i)}$$

- Dirichlet is the multivariate generalization of the Beta distribution.

Poisson-Gamma Conjugacy

- Likelihood:

$$x_n \sim \text{Poisson}(\lambda)$$

- Prior:

$$\lambda \sim \text{Gamma}(a, b)$$

- Posterior:

$$\lambda|\mathcal{D} \sim \text{Gamma}(a + \sum x_n, b + N)$$

- Posterior mean:

$$\mathbb{E}[\lambda|\mathcal{D}] = \frac{a + \sum x_n}{b + N}$$

- Again, the prior can be interpreted in terms of **pseudo-observations**.
- Gamma is also conjugate to the exponential likelihood.

Exponential Family Form

- Many common likelihoods belong to the **exponential family**, which allows us to define conjugate priors systematically.

- Exponential family form:

$$p(x|\theta) = h(x)\exp(\eta(\theta)^\top T(x) - A(\theta))$$

- For a conjugate prior, we define:

$$p(\theta) \propto \exp(\eta(\theta)^\top \tau - \nu A(\theta))$$

- where:

- τ are the prior pseudo-observations (sufficient statistics)
- ν is the effective sample size

- Posterior has the same form:

- posterior:

$$p(\theta|\mathcal{D}) \propto \exp\left(\eta(\theta)^\top (\tau + \sum T(x_n)) - (\nu + N)A(\theta)\right)$$

- The exponential family makes it easy to generalize conjugate priors across many models.

Benefits of Conjugate Priors

- **Closed-form posterior:** No need for sampling or approximation.
- **Ease of computation:** Simple updates using sufficient statistics.
- **Interpretability:** Posterior parameters have intuitive meanings (e.g., prior counts).
- **Online updates:** Easy to incorporate new data without redoing inference from scratch.
- Conjugacy is not always realistic, but it is analytically elegant and computationally efficient.

Limitations of Conjugate Priors

- May not reflect true prior knowledge:
 - E.g., beta prior implies continuous support on $\theta \in (0,1)$ even if domain knowledge suggests otherwise.
- Can be overly simplistic in complex models.
- Lack of flexibility: cannot model multimodal or bounded beliefs well.
- Not all likelihoods have known conjugate priors.
- Real-world applications may require sacrificing conjugacy for realism.

Cf. negative transfer

Noninformative Priors

What Are Noninformative Priors?

- **Goal:** Use priors that have minimal influence on the posterior.
- Often used when we lack strong prior knowledge.
- Also called: *objective*, *reference*, or *uninformative* priors.
- Common in classical statistics (e.g., Jeffreys prior).
- Plays a role in **default Bayesian analysis**.

- In practice, one must still be careful, "uninformative" doesn't mean "safe."

Uniform Priors

- Uniform prior: $p(\theta) \propto 1$
- Example: $\theta \sim \mathcal{U}(a, b)$
- Appears “non-informative” but:
 - Improper if $a = -\infty, b = \infty$
 - Sensitive to reparameterization
- Problem: what looks flat in one parameterization may not in another.
- Uniform in θ is not uniform in $\log\theta$.

Invariance and the Problem with Uniform Priors

- Consider change of variables: $\theta = f(\phi)$
- If $p(\theta) \propto 1$, then

$$p(\phi) \propto \left| \frac{d\theta}{d\phi} \right|$$

- Hence, uniform prior depends on parameterization.
- This breaks **reparameterization invariance**.
- We need a way to define priors invariant to reparameterization.

Jeffreys Prior

- Proposed by Harold Jeffreys.
- Defined as:

$$p(\theta) \propto \sqrt{I(\theta)}$$

- where $I(\theta)$ is the Fisher information:

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p(x|\theta) \right)^2 \right]$$

- Invariant under reparameterization.
- Considered a principled choice for noninformative prior.

Example – Bernoulli Likelihood

- Model: $x \sim \text{Bernoulli}(\theta)$

- Likelihood:

$$p(x|\theta) = \theta^x (1 - \theta)^{1-x}$$

- Fisher information:

$$I(\theta) = \frac{1}{\theta(1 - \theta)}$$

- Jeffreys prior:

$$p(\theta) \propto \sqrt{\frac{1}{\theta(1 - \theta)}}$$

- This is a **Beta(0.5, 0.5)** distribution.
- Stronger belief in values near 0 or 1 (U-shaped).

Example – Gaussian with Known Variance

- Model: $x \sim \mathcal{N}(\mu, \sigma^2)$, with known σ^2
- Fisher information:

$$I(\mu) = \frac{1}{\sigma^2}$$

- Jeffreys prior:

$$p(\mu) \propto \sqrt{\frac{1}{\sigma^2}} \Rightarrow p(\mu) \propto 1$$

- Hence Jeffreys prior is flat in this case.
- Flat prior is justified when variance is fixed.

Example – Gaussian with Unknown Variance

- Model: $x \sim \mathcal{N}(\mu, \sigma^2)$, both unknown
- Parameters:

$$\theta = (\mu, \sigma^2)$$

- Fisher information matrix:

$$I(\theta) = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}$$

- Jeffreys prior:

$$p(\mu, \sigma^2) \propto \sqrt{|I(\theta)|} = \frac{1}{\sigma^2}$$

- Matches conjugate prior form: improper scale-invariant prior.

Improper Priors

- A prior is **improper** if it doesn't integrate to 1:

$$\int p(\theta) d\theta = \infty$$

- Examples:
 - $p(\mu) \propto 1$ over $\mathbb{R}$
 - $p(\sigma^2) \propto 1/\sigma^2$ over $\mathbb{R}^+$
- Improper priors can still yield **proper posteriors**.
- Improper priors can cause problems in some models (e.g., hierarchical models).

Hierarchical Bayes

Motivation for Hierarchical Bayes

- Many real-world problems involve *grouped* or *multi-level* data.
- Hierarchical Bayes allows **sharing statistical strength** across groups.
- Enables **modeling variability across groups** while maintaining a common prior structure.
- Examples:
 - School test scores from different cities
 - Hospital recovery rates across regions
- “Borrowing strength” from other data groups improves estimation when individual groups are small.

What is a Hierarchical Prior?

- A **hierarchical prior** introduces another layer of uncertainty.
- Parameters themselves have distributions with hyperparameters.
- These hyperparameters may also have priors (hyperpriors).
- Helps capture parameter uncertainty and improve regularization.
- General structure:

$$\begin{aligned}\theta &\sim p(\theta|\lambda) \\ \lambda &\sim p(\lambda)\end{aligned}$$

Another Motivation for Hierarchical Priors

- Reduces sensitivity to specific hyperparameter choices.
- Introduces **adaptive shrinkage**, data can inform regularization strength.
- More **robust** modeling: especially useful in high dimensions or sparse data.
- Enables **sharing of statistical strength** across related parameters (e.g., multi-task learning).
- Encourages “soft tying” among model parameters.

Example - Gaussian Prior with Hierarchical Variance

- Likelihood:

$$x_n \sim \mathcal{N}(\theta, \sigma^2)$$

- Prior:

$$\theta \sim \mathcal{N}(0, \lambda^{-1})$$

- Hyperprior:

$$\lambda \sim \text{Gamma}(a, b)$$

- Posterior over (θ) becomes more flexible by adapting (λ) .
- Gamma hyperprior controls the amount of shrinkage.

Posterior Computation in Hierarchical Models

- Posterior is:

$$p(\theta, \lambda | \mathcal{D}) \propto p(\mathcal{D} | \theta) p(\theta | \lambda) p(\lambda)$$

- Often no closed-form, especially when marginalizing over λ :

$$p(\theta | \mathcal{D}) = \int p(\theta | \lambda, \mathcal{D}) p(\lambda | \mathcal{D}) d\lambda$$

- Approximate inference required (MCMC or variational methods).
- Integrating out λ leads to heavier-tailed posteriors.

Structure of a Hierarchical Model

- Two or more levels of parameters:
 - Level 1 (local): group-specific parameters θ_k
 - Level 2 (global): shared hyperparameters α
- Generative process:
 - Sample global hyperparameters: $\alpha \sim p(\alpha)$
 - For each group k :
 - $\theta_k \sim p(\theta_k|\alpha)$
 - $x_{nk} \sim p(x_{nk}|\theta_k)$
- Posterior is typically intractable $\rightarrow$ need approximate inference (e.g. MCMC).

Example - Gaussian-Gaussian Hierarchical Model

- Observation model:

$$x_{nk} \sim \mathcal{N}(\mu_k, \sigma^2)$$

- Prior for each mean:

$$\mu_k \sim \mathcal{N}(\mu_0, \tau^2)$$

- Hyperprior for μ_0 :

$$\mu_0 \sim \mathcal{N}(m, s^2)$$

- Resulting posterior:

$$p(\mu_0, \{\mu_k\}|\mathcal{D}) \propto p(\mu_0) \prod_k p(\mu_k|\mu_0) \prod_n p(x_{nk}|\mu_k)$$

- Posterior over all parameters is not tractable in closed-form.

Shrinkage Effect in Hierarchical Models

- MAP estimate for μ_k is a **compromise**:

$$\hat{\mu}_k = \lambda \bar{x}_k + (1 - \lambda) \mu_0$$

- where:
 - $\bar{x}_k$ is empirical mean in group k
 - μ_0 is global mean
 - $\lambda = \frac{\tau^2}{\tau^2 + \sigma^2 / N_k}$
- Small group size $N_k \Rightarrow$ more shrinkage to μ_0
- Large group $\Rightarrow$ more influence from $\bar{x}_k$
- Automatic regularization via Bayesian hierarchy.

Hierarchical Prior as Implicit Regularizer

- Hierarchical models provide **adaptive regularization**:
 - Data-poor groups: rely on global prior
 - Data-rich groups: rely on own data
- Similar to ridge regression, but **Bayesian and probabilistic**
- Especially useful in high-dimensional or sparse group settings

Posterior Inference Challenges

- Posterior distribution:
- $p(\theta_{1:K}, \alpha | \mathcal{D}) \propto p(\alpha) \prod_k p(\theta_k | \alpha) \prod_n p(x_{nk} | \theta_k)$
- No closed-form solution in general
- Inference techniques:
 - Gibbs sampling or Metropolis-Hastings (MCMC)
 - Variational Bayes approximations
 - Empirical Bayes for estimating α via marginal likelihood
- Realistic but computationally intensive.

Empirical Bayes

What is Empirical Bayes?

- Definition

- Empirical Bayes (EB) is a hybrid approach where the prior distribution's hyperparameters are estimated from data rather than fixed subjectively.
- Instead of fully Bayesian integration over hyperparameters, EB plugs in point estimates.

- Motivation

- Priors are often hard to elicit from domain experts.
- Large datasets provide enough information to estimate priors directly.
- A hybrid approach between Bayesian and frequentist inference.

- Rather than placing a prior $p(\theta|\alpha)$ and integrating over α ,

- Estimate α from data using MLE or MAP.
- Plug-in the estimate into the prior.

- Resulting posterior:

$$p(\theta|\mathcal{D}, \hat{\alpha}) \propto p(\mathcal{D}|\theta)p(\theta|\hat{\alpha})$$

Type-II Maximum Likelihood

- Recall Bayes rule:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

- In EB:

- Estimate prior hyperparameters ϕ from data:

$$\hat{\phi} = \arg \max_{\phi} p(\mathcal{D} \mid \phi)$$

- Plug-in approach:

$$p(\theta \mid \mathcal{D}) \approx p(\theta \mid \mathcal{D}, \hat{\phi})$$

- Prior is **empirical**, learned from data, not fixed in advance.

Marginal Likelihood for Hyperparameter Estimation

- Marginal likelihood:

$$p(\mathcal{D} \mid \phi) = \int p(\mathcal{D} \mid \theta) p(\theta \mid \phi) d\theta$$

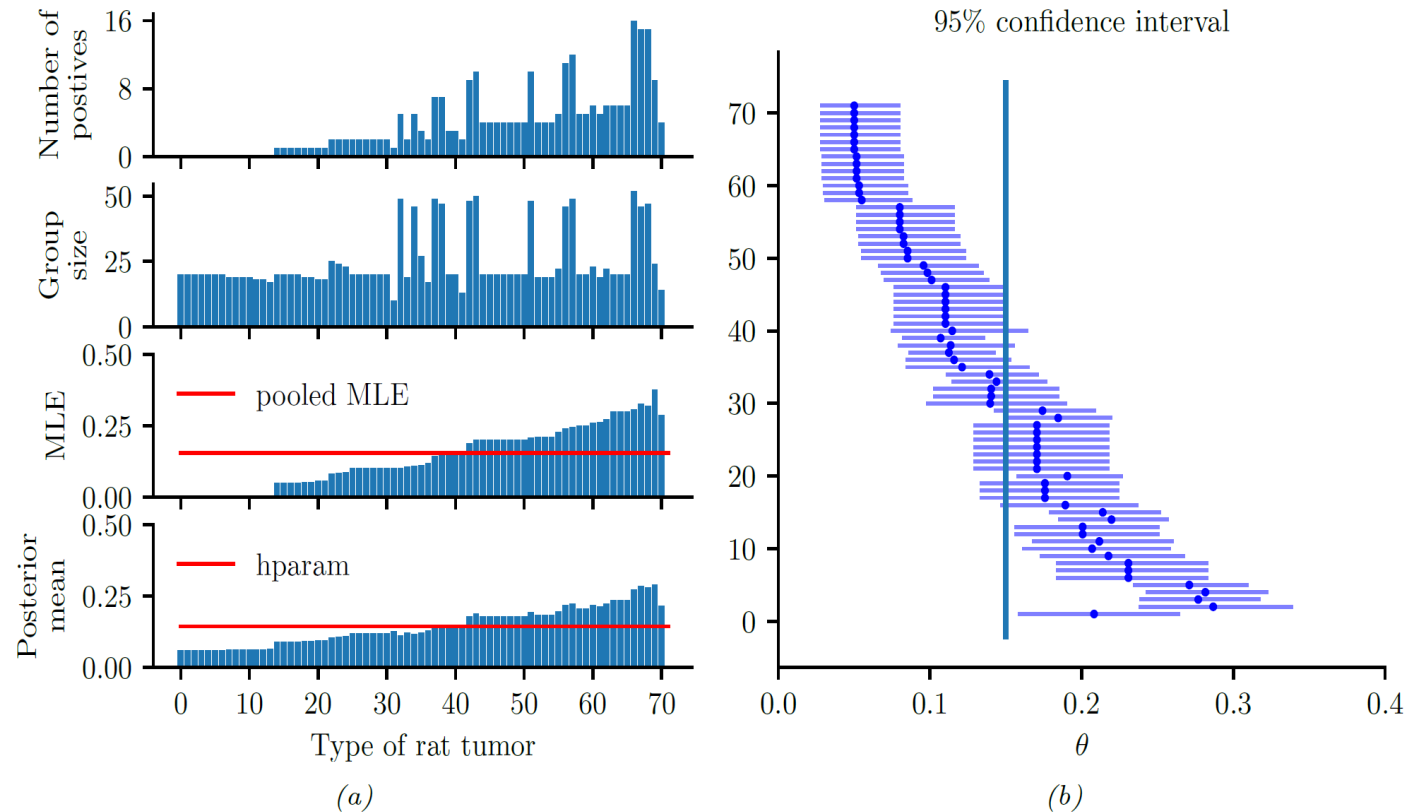
- We optimize this to find $\hat{\phi}$

$$\hat{\phi} = \arg \max_{\phi} p(\mathcal{D} \mid \phi)$$

- Often intractable integral $\rightarrow$ requires approximation.
 - Laplace approximation
 - EM algorithm (e.g., for latent variable models)
 - Monte Carlo methods
- Once $\hat{\phi}$ is obtained,
 - we compute: $p(\theta \mid \mathcal{D}) \approx p(\theta \mid \mathcal{D}, \hat{\phi})$

EB for the Hierarchical Binomial Model

- **Objective** - Fit a hierarchical binomial model using Empirical Bayes (EB), avoiding full Bayesian inference. - Use marginal likelihood optimization to estimate hyperparameters (a) and (b).



EB for the Hierarchical Binomial Model

- Suppose: $y_i \sim \text{Binomial}(n_i, \theta_i), \theta_i \sim \text{Beta}(\alpha, \beta)$
- We can analytically integrate out the θ_j 's, and write down the marginal likelihood directly
- **Marginal Likelihood Expression From the model:**

$$\begin{aligned} p(\mathcal{D}|\xi) &= \prod_j \int \text{Bin}(y_j|N_j, \theta_j) \text{Beta}(\theta_j|a, b) d\theta_j \\ &\propto \prod_j \frac{B(a + y_j, b + N_j - y_j)}{B(a, b)} \\ &= \prod_j \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a + y_j)\Gamma(b + N_j - y_j)}{\Gamma(a + b + N_j)} \end{aligned}$$

- Closed-form via Beta-Binomial conjugacy: $= \prod_{i=1}^N \text{BetaBin}(y_i | N_i, \alpha, \beta)$
- Estimate $\hat{\alpha}, \hat{\beta}$, then: $p(y_i | N_i, \hat{\alpha}, \hat{\beta}) = \text{Beta}(y_i + \hat{\alpha} | N_i - \hat{\alpha} + \beta)$

Example 1 – Gaussian Mean Estimation

- Data: $x_i \sim \mathcal{N}(\theta_i, \sigma^2)$
- Prior: $\theta_i \sim \mathcal{N}(\mu, \tau^2)$
- Hyperparams: $\alpha = (\mu, \tau^2)$
- Estimate $\hat{\mu}, \hat{\tau}^2$ from $\{x_i\}$ via MLE:
- $\hat{\mu} = \frac{1}{N} \sum x_i, \quad \hat{\tau}^2 = \max(0, s^2 - \sigma^2)$
- Posterior of θ_i given x_i and $\hat{\alpha}$:
- $p(\theta_i | x_i, \hat{\alpha}) \sim \mathcal{N}(m_i, v_i)$
- Closed-form expressions for m_i, v_i available.
- Illustrates shrinkage: posterior mean m_i is shrunk toward $\hat{\mu}$.

Example 2 – Gamma-Poisson Model

- Data: $x_i \sim \text{Poisson}(\theta_i)$
- Prior: $\theta_i \sim \text{Gamma}(a, b)$
- Estimate a, b from data:
- $\hat{a}, \hat{b} = \underset{a, b}{\operatorname{argmax}} \prod_{i=1}^N \int \text{Poisson}(x_i | \theta_i) \text{Gamma}(\theta_i | a, b) d\theta_i$
- Closed-form expression exists via marginal likelihood:
- $p(x_i | a, b) = \text{NegBin}(x_i | a, b/(1 + b))$
- Then compute posterior:
- $p(\theta_i | x_i, \hat{a}, \hat{b}) \sim \text{Gamma}(a + x_i, b + 1)$

Benefits of Empirical Bayes

- **Scalability:** Fast for large number of units (e.g., documents, users).
- **Automation:** Prior learned from data — less manual tuning.
- **Interpretability:** Estimated prior gives insight into population-level behavior.
- Appealing when **many repeated sub-problems** share prior info
- Common in:
 - Hierarchical models
 - Genomics
 - Natural language processing
 - Sports analytics (e.g., batting averages)
- Shrinkage estimates: EB helps regularize over noisy data.

Limitations of Empirical Bayes

- Underestimates posterior variance (treats $\hat{\alpha}$ as fixed)
- Not truly Bayesian (doesn't integrate over uncertainty in α)
- Sensitive to MLE errors if small data
- Can lead to **over-shrinkage** or miscalibrated uncertainty
- Less robust than full Bayes in high-variance settings
- Always validate EB assumptions; consider full Bayes if uncertainty is critical.

Model Selection

What is Model Selection?

- **Goal:** Choose the best model M_k from a set of candidates $\{M_1, M_2, \dots, M_K\}$
- **Why it matters:**
 - Prevent overfitting.
 - Improve generalization.
 - Quantify trade-off between model complexity and fit.
- **Each model M_k has:**
 - Its own parameter space Θ_k
 - Prior $p(\theta_k | M_k)$
 - Likelihood $p(\mathcal{D} | \theta_k, M_k)$
- In Bayesian setting, we compute:

$$p(M_k | \mathcal{D}) \propto p(\mathcal{D} | M_k) p(M_k)$$

- where:

$$p(\mathcal{D} | M_k) = \int p(\mathcal{D} | \theta_k, M_k) p(\theta_k | M_k) d\theta_k$$

Prior and Posterior over Models

- **Model prior:** $p(M_k)$ expresses our belief about model plausibility before seeing data.

- **Posterior over models:**

$$p(M_k | \mathcal{D}) = \frac{p(\mathcal{D} | M_k) p(M_k)}{\sum_j p(\mathcal{D} | M_j) p(M_j)}$$

- Called **Bayesian Model Averaging (BMA)** denominator
- Used in:
 - Model comparison
 - Model averaging
 - Ensemble predictions

Marginal Likelihood (Model Evidence)

- Key quantity in Bayesian model selection:

$$p(\mathcal{D} \mid M_k) = \int p(\mathcal{D} \mid \theta_k, M_k) p(\theta_k \mid M_k) d\theta_k$$

- **Occam's Razor effect:**
 - Penalizes complex models unless justified by the data.
 - Prefers models that balance fit and simplicity.
- This term automatically trades off goodness-of-fit and model complexity.

Example – Coin Toss with Two Models

- **Model 1:** Fixed coin bias $\theta = 0.5$
- **Model 2:** Unknown bias $\theta \sim \text{Beta}(a, b)$
- Compute:
 - $p(\mathcal{D} \mid M_1) = \prod_n 0.5^{x_n} (1 - 0.5)^{1-x_n}$
 - $p(\mathcal{D} \mid M_2) = \int p(\mathcal{D} \mid \theta) p(\theta) d\theta$
- Marginal likelihood favors simpler models unless there is strong evidence.

BIC (Bayesian Information Criterion)

- Approximate marginal likelihood:

$$\log p(\mathcal{D} \mid M_k) \approx \log p(\mathcal{D} \mid \hat{\theta}_k) - \frac{d_k}{2} \log N$$

- where:

- $\hat{\theta}_k$:MLE estimate
- d_k :number of parameters
- N : number of data points

- **Interpretation:**

- Fit term: $\log p(\mathcal{D} \mid \hat{\theta}_k)$
- Penalty term: $\frac{d_k}{2} \log N$

- BIC favors models with high likelihood and fewer parameters.

AIC (Akaike Information Criterion)

- Based on predictive accuracy (frequentist):

$$\text{AIC} = -2\log p(\mathcal{D}|\hat{\theta}) + 2d$$

- Lower AIC is better.
- Does not depend on prior.
- AIC $\neq$ BIC: different goals - AIC: minimize predictive error - BIC: maximize model posterior

Cross-Validation for Model Selection

- Split data into K folds.
- For each fold:
 - Train on $K-1$ folds
 - Test on 1 fold
 - Compute average predictive performance.
- Common metrics: log-likelihood, RMSE, accuracy.
- *Pros:*
 - No need for prior.
 - Works for any model.
- *Cons:*
 - Computationally expensive
 - No posterior uncertainty over models

Model Averaging vs Model Selection

- **Model Selection:**

- Choose $\arg \max_k p(M_k | \mathcal{D})$
- Risk: underestimating uncertainty

- **Model Averaging:**

- Predictive distribution:

$$p(y | \mathcal{D}) = \sum_k p(y | M_k, \mathcal{D}) p(M_k | \mathcal{D})$$

- Weighs models by posterior probability
- Averaging accounts for model uncertainty and often improves predictions.

Bayes Factor vs p-value

- p-values overstate evidence against H_0 .
- Bayes factors offer more calibrated interpretation.

Feature	Bayes Factor	p-value
Interpretation	Data under H_1 vs H_0	Probability under H_0
Model Comparison	Yes	No
Penalizes Complexity	Yes (via prior)	No
Evidence for H_0	Possible	No (can only reject H_0)
Prior Needed	Yes	No

Q & A

References

- Murphy, K. P. (2023). *Probabilistic machine learning: Advanced topics*. MIT press.